

Sujay Seeram

Software Systems, AI Infrastructure & High-Performance Runtimes Engineer

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ENGINEERING PROFILE & CORE PHILOSOPHY

Undergraduate systems and AI engineer at Amrita Vishwa Vidyapeetham with an established track record in **high-performance distributed runtimes, local-first sovereign AI architectures**, and **physics-informed machine learning**. Pioneer of the “**Extreme Economic Engineering**” paradigm: combining mathematical first principles and algorithmic efficiency to execute complex neural inference, vector search, and interactive streaming on low-cost host hardware without cloud dependency. Disciplined NCC cadet and 3-time Services Selection Board (SSB) attendee.

EDUCATION

- **Amrita Vishwa Vidyapeetham** Coimbatore, India
Bachelor of Technology in Electronics and Communication Engineering (ECE) *Expected 2027*
 - **Relevant Coursework:** Data Structures & Algorithms, Operating Systems Concepts, Computer Networks, Linear Algebra, Probability & Stochastic Processes, Object-Oriented Software Design, Database Management Systems.
 - **Core Focus:** High-Throughput I/O Runtimes, Local Vector Indexing, Physics-Informed ML, Autonomous Agent Protocols.

TECHNICAL SKILLS

- **Programming Languages:** Python, Rust, TypeScript, JavaScript, C/C++17, Go, Dart (Flutter), SQL, POSIX Shell/Bash.
- **Systems, Networking & Runtimes:** Actix-Web, Tokio async, FastAPI, Node.js, WebSockets, PTY orchestration, Linux eBPF, Docker, Linux IPC.
- **Machine Learning, DSP & Retrieval:** PyTorch, Scikit-Learn, LanceDB (Vector Indexing), SQLite Local-RAG, Model Context Protocol (MCP), Gemini API, Fast Fourier Transform (FFT).
- **Frontend & UI Systems:** React 18, Vite, React Native, Three.js / React Three Fiber, WebGL, Reanimated, Streamlit.
- **Cloud, Security & Databases:** Google Cloud Run, Firebase Auth (OIDC), Cloud Firestore, PostgreSQL, Redis, Android Accessibility API Security.

LEADERSHIP, COMPETITIONS & PROFESSIONAL DISTINCTIONS

- **IIT Madras Smart India Challenge – 2nd Runner Up & Youngest Panelist** *IIT Madras*
 - Awarded **2nd Runner Up** at IIT Madras Smart India Challenge for developing high-efficiency distributed and cyber-physical architectures.
 - Invited and served as the **Youngest Panelist** at the summit, presenting on low-overhead systems and sovereign AI computation.
- **Global Automotive Research Centre (GARC) Technical Delegation** *Chennai, India*
 - Selected technical delegate at the **GARC Chennai Meet**, engaging with engineers on automotive embedded standards and telemetry.
 - Participated in sessions on real-time telemetry pipelines, vehicle sensor failure preemption, and edge computing diagnostics.
- **National Cadet Corps (NCC) & Services Selection Board (SSB)** *Armed Forces Selection*
 - Trained cadet in the **National Cadet Corps (NCC)**, instilling strong leadership, tactical problem-solving, and disciplined execution.
 - Recommended for and attended the **Services Selection Board (SSB) 3 times**, demonstrating high resilience and situational decision-making.

- **EVolvAI – Generative AI & Physics-Constrained Grid Optimization** *Release v1.0.0*
 - Architected a scalable cyber-physical demand framework in PyTorch coupling a **Generative Counterfactual VAE (GCD-VAE)** with a Multi-Objective Genetic Algorithm (GA).
 - Generated thousands of non-linear spatial charging scenarios under distribution grid bus limits and severe climate stress anomalies, achieving $R^2 \approx 0.5$ predictive accuracy on sparse field data.
 - Engineered an interactive Streamlit simulation console enabling municipal engineers to audit transformer stress curves.
- **ytstr – High-Performance Psychoacoustic Audio Engine** *Release v2.0.0*
 - Developed a terminal-native streaming runtime powered by a psychoacoustic **DecisionEngine** executing real-time spectral analysis.
 - Implemented 8 broadcast-grade transition algorithms (Bass Swap, Frequency Melt, Tape Stop) and built an active backpressure memory pool.
 - Slashed runtime RAM footprint by **70%** (< 260MB peak) during uninterrupted multi-hour streaming.
- **SpecRAG – Zero-Hallucination Firmware Auditing Gateway** *Release v0.1.0*
 - Engineered an enterprise developer tool bridging 1,000-page silicon datasheets with IDE source code using the **Model Context Protocol (MCP)**.
 - Performed localized vector retrieval against indexed silicon register banks to catch peripheral pin collisions, clock-tree violations, and timing hazards directly in VS Code before compilation.
- **MotorSafe – Tri-Modal Distributed Telemetry Engine & Digital Twin** *Release v0.01*
 - Constructed a real-time telemetry streaming system ingesting 3 synchronized sensor channels over MQTT.
 - Designed localized unsupervised **Isolation Forest** models and Fast Fourier Transform (FFT) DSP pipelines paired with an interactive 3D WebGL/Three.js digital twin.
- **Informed-Poll – Swipeable Civic Platform with LanceDB RAG** *Release v1.0.0*
 - Developed a mobile-first civic intelligence companion featured in the Google Solution Challenge.
 - Grounded candidate data with **LanceDB vector store** and Gemini 3.1 Flash Lite over a containerized FastAPI backend on Google Cloud Run with zero-hallucination verification.
- **Kavach-UPI – Android Accessibility Security Shield** *Repository*
 - Authored an on-device security shield leveraging Android Accessibility APIs to detect active screen-sharing malware during financial UPI transactions.
 - Instantly deployed an opaque system window overlay to block remote touch inputs while triggering localized audio warnings.
- **ScopeCreep.ai – LLM Watchdog for Development Integrity** *Repository*
 - Created an automated CI/CD monitoring agent evaluating pull requests and commit diffs against Product Requirement Documents (PRDs).
 - Flagged unauthorized scope inflation and gold-plating in real time before sprint release branches were merged.
- **TurnBack (Activity-Mapper) – Local-First Endurance Tracker** *Release v1.3.0*
 - Built a local-first GPS activity tracking app with a 54% fatigue-asymmetric turn-back navigation engine.
 - Engineered stationary stop detection and Ramer-Douglas-Peucker (RDP) path decimation, maintaining 60 FPS rendering under < 5MB RAM.